Yessenia Nicacio

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Professional Summary

Entry-level Computer Engineer with hands-on experience designing, integrating, validating embedded and data acquisition systems in lab and field environments. Skilled in hardware troubleshooting, system-level testing, and technical documentation. Experienced with collaborating across multidisciplinary teams to deliver hardware and software solutions. Seeking an engineering role in embedded systems, application engineering, hardware validation, test engineering, or technical support.

Technical Skills

Hardware & Embedded: Raspberry Pi, Sensor integration, Embedded systems Debugging, Field Deployment Systems
Tools: Linux, Git, GitHub, VS Code, Microsoft Office, Vivado, Docker, MATLAB
Programming (Familiar): Python, C/C++, Verilog
Instrumentation Exposure: Oscilloscope, Signal Generators, Power Supplies, Basic RF measurement Concepts

Projects

Grainger Computing Innovation Prize - SeismoAI
Founder, Team Lead & Developer Aug. 2025 – Nov.2025

- Designed and validated a portable data acquisition and signal processing system
- Implemented real-time waveform analysis and automated event detection using a python-based AI pipeline
- Performed system-level testing and instrumentation troubleshooting to resolve sensor connectivity issues
- Developed technical documentation detailing system architecture and testing procedures
- Led cross-disciplinary team between hardware, software and business
- Awarded **2nd place** at the Grainger Computing Innovation Competition, earning **$10,000** in prize funding.

J.A.R.V.I.S. – An interactive Everyday Assistant (Senior Capstone Project)
Mobile App & Frontend Developer Aug.2025 – Dec.2025

- Integrated a mobile application with Raspberry Pi–based hardware using MQTT and communication to enable system control
- Tested and validated application-to-hardware messaging, troubleshooting connectivity and broker configuration issues
- Documented communication workflows, system configuration, and testing results for final project evaluation

Professional Experience

Sigenics Inc | Chicago, IL
Seasonal Engineering Assistant May 2025 – Aug. 2025

- Utilized industry-standard layout analysis software to review semiconductor cross-section for verification support
- Developed Digital communication interface diagrams to document signal flow architectures for integrated circuit development
- Supported semiconductor lab operations including device handling, organization and packaging in controlled environment
- Assisted in improving technical data organization and workflow efficiency

University of Illinois at Urbana-Champaign | Prairie Research Institute
Research Fellow Intern May 2024 – July 2024

- Configured and maintained Raspberry Pi based seismic measurement system for reliable field data acquisition
- Developed Python Scripts for signal processing and data visualization
- Performed hardware validation and troubleshooting to ensure consistent sensor performance
- Delivered technical reports and communicated findings to research team

University of Illinois at Urbana-Champaign | Prairie Research Institute
SROP Research Intern July 2023 – August 2023

- Installed a Raspberry Shake seismometer to replace a commercial system at the National Sequestration Education Center.
- Analyzed waveform data using Waves software for carbon-sequestration monitoring.
- Presented research at the 2023 Illinois Summer Research Symposium:"Enhancing Community Confidence in Geologic Carbon Sequestration through Raspberry-Shake and Open Data.

Education

Illinois Institute of Technology (Illinois Tech) - Chicago, IL Sept. 2023 – Dec. 2025
Bachelor of Science in Computer Engineering

Wilbur Wright College – Chicago, IL Jan. 2020 – Sept. 2024
Associates - Engineering Science Electrical Engineering

Leadership & Research

Society of Hispanic Professional Engineers (SHPE) – President Sept.2022 – June 2023

- Led a 15-member executive board, coordinating organizational planning, events, and professional development activities
- Increased chapter membership through outreach initiatives and technical workshops

NSF HSI (Hispanic- Serving Institutions) Research Assistant Oct. 2022 – July 2023

- Co-authored an ASEE-published paper on STEM identity and student engagement.
- Presented research findings at the 2023 ASEE National Conference in Baltimore